A Complete Guide to Vector Databases

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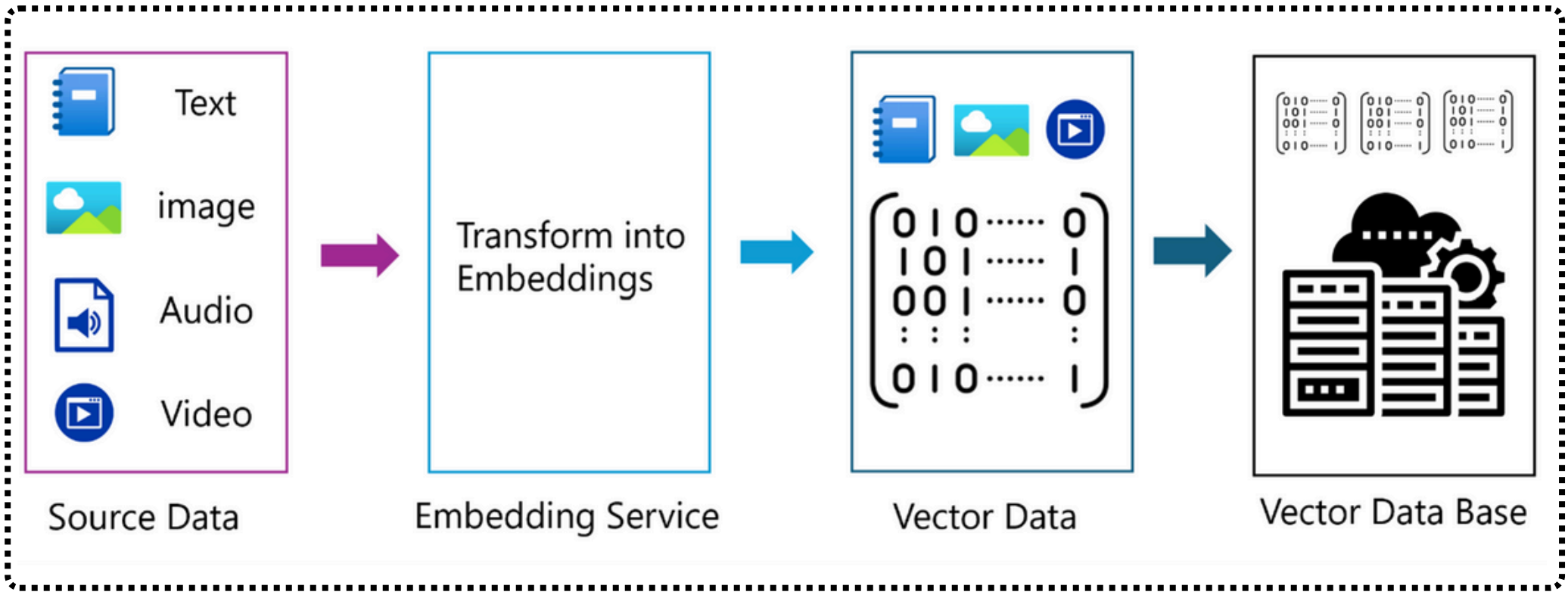
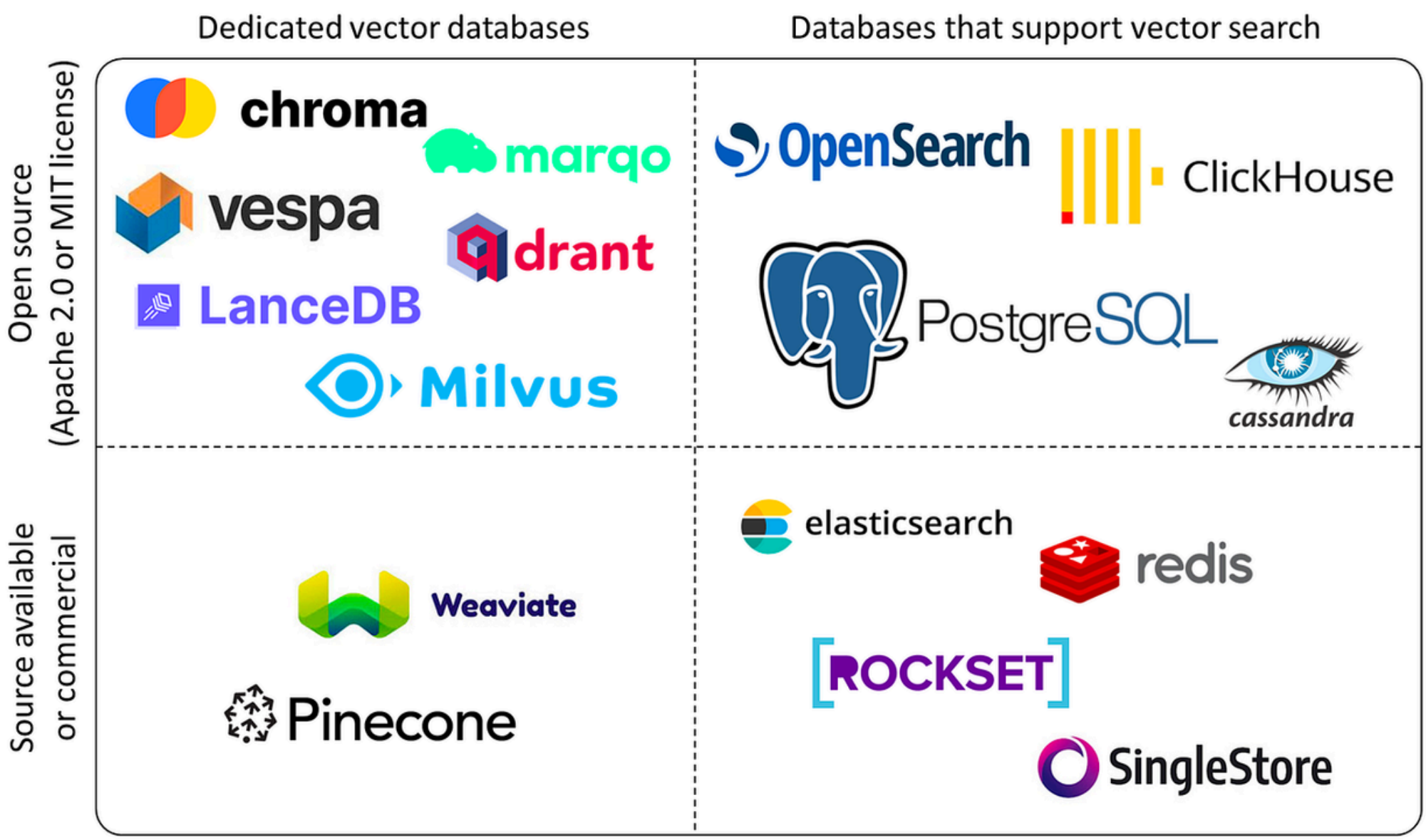
Complete Notes





1. Why Vector Databases Exist

Traditional databases are good at answering questions like:

- Find users with age = 30
- Find orders where price > 100
- Find products where category = shoes

But they completely fail at questions like:

- Find text similar to “user is angry about late delivery”
- Find images similar to this photo
- Find documents related to “LLM hallucination”
- Find code similar to this function

These are **semantic** queries.

They are about **meaning**, not exact values.

That is why **vector databases exist**.

They store data as **vectors** so that **similar things are close together** in mathematical space.

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2. What is a Vector?

A **vector** is just a list of numbers.

Example:

csharp Copy code

```csharp
[0.21, -0.98, 1.34, 0.07, ...]
```

In ML, vectors represent meaning.

Text, images, audio, code all get converted into vectors using **embedding models**.

Data	Embedding
"I love dogs"	[0.12, -0.44, 1.2, ...]
"I like puppies"	[0.11, -0.46, 1.18, ...]
"I hate traffic"	[-0.91, 0.02, -1.4, ...]

Notice how similar meanings produce similar vectors.

3. What is a Vector Database?

A **vector database** stores:

- Vectors
- Metadata
- Original content

and supports:

- Fast similarity search
- Filtering
- Scalable retrieval

Instead of SQL like:

```java
WHERE age = 30
```

You query using:

```cpp
Find top 10 vectors closest to this vector
```

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4. What Makes Vector Search Work

Vectors live in a high dimensional space.

A text embedding might have 768 or 1536 dimensions.

Similarity is computed using distance functions:

Metric	Meaning
Cosine similarity	Angle between vectors
Euclidean distance	Straight line distance
Dot product	Directional similarity

Cosine similarity is most common for text.

5. Why You Cannot Use SQL for This

SQL databases:

- Are row based
- Use indexes on fixed columns
- Work with exact values

Vector search requires:

- High dimensional nearest neighbor search
- Approximate search
- In memory optimized indexes

Trying this in PostgreSQL or MySQL will be slow and painful.

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6. Core Architecture of a Vector Database

A vector DB has 4 main layers.

1. Ingestion Layer

Takes:

- Raw data
- Vectors
- Metadata

2. Indexing Layer

Builds ANN indexes like:

- HNSW
- IVF
- PQ

This allows fast similarity search.

3. Storage Layer

Stores:

- Vectors
- Metadata
- IDs

4. Query Engine

Accepts:

- A vector
- Filters
- Top K

Returns:

- Most similar items

7. ANN Search Explained Simply

ANN means **Approximate Nearest Neighbor**.

Exact search checks every vector.
That is impossible at scale.

ANN uses graph and clustering tricks to search only the promising part of space.

HNSW builds a layered graph where similar vectors connect.

IVF clusters vectors and only searches relevant clusters.

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8. How Data Flows

Let's say you want to build an AI document search.

Step 1: Chunk your documents

Split PDFs, docs, pages into chunks.

Step 2: Convert text to vectors

Using an embedding model.

```arduino
"This product is bad" → [0.34, -1.2, ...]
```

Step 3: Store in vector DB

You store:

```bash
id: 1
vector: [0.34, -1.2, ...]
text: "This product is bad"
metadata: {source: "review", date: "2024-01-01"}
```

Step 4: User asks a question

"Why are customers unhappy?"

Convert the question into a vector.

Step 5: Similarity search

Find nearest vectors.

Step 6: Send results to LLM

Now the LLM answers using real data.

This is called **RAG**.

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9. Why Vector DBs Power LLMs

LLMs do not remember your data.

Vector databases give LLMs memory.

This enables:

- Chat with PDFs
- Customer support bots
- Enterprise search
- Code assistants
- Recommendation systems

Without vector DBs, LLMs hallucinate.

10. Popular Vector Databases

Name	Notes
Pinecone	Fully managed
Weaviate	Open source + cloud
Milvus	High scale open source
Qdrant	Fast, Rust based
FAISS	Library, not DB
Chroma	Local dev
pgvector	Postgres plugin

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11. What Makes a Good Vector Database

You want:

- Fast ANN search
- Filtering on metadata
- Horizontal scaling
- Persistence
- Hybrid search (text + vectors)
- High recall

12. Vector DB vs Search Engine

Search Engine	Vector Database
Keyword based	Meaning based
BM25	Embeddings
Exact matches	Semantic matches
Misses paraphrases	Handles synonyms

Modern systems combine both.

13. Hybrid Search

Best systems do:

```nginx
vector_score * 0.7 + keyword_score * 0.3
```

This gives accuracy and semantic power.

14. Cost Model

Vector DBs are expensive because:

- Vectors are large
- RAM heavy
- Indexing is complex

Cost drivers:

- Number of vectors
- Dimension size
- Replication
- Query volume

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15. When You Should Use One

Use a vector DB if you need:

- Semantic search
- Recommendation engines
- AI agents with memory
- Document QA
- Similarity matching
- Fraud detection
- Image or audio search

Do not use one for simple CRUD.

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16. The Mental Model

Think of a vector DB as:

A Google Maps of meaning.

Queries are locations.
Data points are places.
Nearby means similar.